

4.00-hpi-german-cc-example_dirac

August 30, 2024

1 PRiSM Example: German Credit Card Data

This notebook demonstrates the PRiSM (Partial Responses in Structured Models) method, which is designed to transform black-box classifiers into inherently interpretable models without compromising predictive performance. This approach is particularly valuable in high-stakes applications where understanding the model’s decision-making process is crucial.

The method addresses a key challenge in machine learning: the trade-off between model complexity and interpretability. It allows us to create models that are both accurate and transparent. In this example, we’ll use a German credit card dataset to illustrate the PRiSM method. While this isn’t a medical dataset, the principles demonstrated here are directly applicable to clinical data and decision support systems.

Here, the target outcome we are modelling is “Class label”, which refers to:

- 0: a customer who **did not default** on their loan
- 1: a customer who **did default** on their loan

We would like to use the available features to model the likelihood of a customer defaulting, and understand how each feature contributes to that likelihood.

1.1 1. Setup and Imports

First, we need to import the necessary libraries and set up our environment.

```
[ ]: %load_ext autoreload
      %autoreload 2

import numpy as np
import pandas as pd
import torch
import matplotlib.pyplot as plt
import seaborn as sns

from prism.config import RAW_DATA_DIR, MODELS_DIR
from prism.preprocessing import normalize, feature_summary, \
    plot_feature_histograms
from prism.maskedmlp import train_mlp_batched, train_maskedmlp
from prism.save_models import save_mlp
from prism.partial_responses import partial_responses
```

```

from prism.response_plot import plot_partial_responses
from prism.nomogram import nomogram, save_nomogram_data_to_csv, \
    display_nomograms_side_by_side
from prism.lasso import lasso
from prism.device_tools import get_device, get_num_cpu_workers
from prism.metrics import evaluate_model_performance, compare_model_performance

```

The autoreload extension is already loaded. To reload it, use:

```
%reload_ext autoreload
```

```
[ ]: %reload_ext autoreload
```

We set up our computational device (CPU or GPU if available) and set a random seed for reproducibility.

We'll use the 'lebesgue' method for partial responses, which is one of two methods (the other being 'dirac') for calculating partial responses. The Lebesgue method uses an average over the predicted surface, which can be more stable for some datasets, but more computationally intensive. If you're working with a bigger dataset, or low compute resources, starting with 'dirac' may be better.

```

[ ]: # Set device to 'cpu', or GPU ('cuda' for NVIDIA, 'mps' for Apple Silicon)
     # By default, we select the best available device
     device = 'cpu' #get_device()
     print(f"Using device: {device}")

     # Set random seed for reproducibility
     random_seed = 257
     np.random.seed(random_seed)
     torch.manual_seed(random_seed)

     # Set method and other parameters
     partial_response_method = 'dirac'
     SAVE_MODELS = False
     SAVE_METRICS = True

```

Using device: cpu

For multithreading compute-intensive tasks, the recommended number of workers for GPU processing is 1, otherwise for CPU stick to the default (number of logical cores - 1). However, depending on your hardware, you may want to experiment.

```

[ ]: if device != 'cpu':
     max_workers = 1
     else:
         max_workers = get_num_cpu_workers()
     print(f"max_workers = {max_workers}")

```

```
max_workers = 15
```

1.2 2. Load and Preprocess Data

Next, we load the German credit card data and perform necessary preprocessing steps.

```
[ ]: from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.impute import SimpleImputer

def preprocess_german_credit_data(data, target_column, feature_columns,
    test_size=0.2, random_state=257, shuffle=False, credit_scale=0.01):
    """
    Preprocess the German Credit Card dataset.

    Parameters:
    data (pd.DataFrame): The raw German Credit Card dataset
    target_column (str or int): The name or index of the target column
    feature_columns (list): List of column names or indices to use as features
    test_size (float): Proportion of the dataset to include in the test split
    random_state (int): Random state for reproducibility

    Returns:
    tuple: X_train, X_test, y_train, y_test (all as pd.DataFrame or pd.Series)
    """
    # Select features and target
    if isinstance(target_column, int):
        target_column = data.columns[target_column]

    if all(isinstance(col, int) for col in feature_columns):
        feature_columns = data.columns[feature_columns]

    X = data.copy()[feature_columns]
    y = data[target_column]

    # Scale the 'Credit amount' if it exists in the dataset
    if 'Credit amount' in data.columns:
        X['Credit amount'] = X['Credit amount'] * credit_scale
        X['Credit amount'] = X['Credit amount'].round()

    # Encode categorical variables
    label_encoder = LabelEncoder()
    for column in X.select_dtypes(include=['object']):
        X[column] = label_encoder.fit_transform(X[column].astype(str))

    # Impute missing values
    imputer = SimpleImputer(strategy='median')
    X = pd.DataFrame(imputer.fit_transform(X), columns=X.columns)

    # Split the data into training and testing sets
```

```

X_train, X_test, y_train, y_test = train_test_split(X, y,
↳test_size=test_size,random_state=random_seed,shuffle=shuffle)

# Ensure the target variable is binary
y_train = (y_train == y_train.max()).astype(int)
y_test = (y_test == y_test.max()).astype(int)

return X_train, X_test, y_train, y_test

```

```

[ ]: # Load data
german_credit_data = pd.read_csv(RAW_DATA_DIR.joinpath('GermanCC_attributes.
↳csv'))

target_column = 'Class label'
feature_columns = [0, 1, 2, 4, 5, 6, 13, 19, 20, 21] # List of column indices
↳to use as features

X_train, X_test, y_train, y_test = preprocess_german_credit_data(
    german_credit_data,
    target_column,
    feature_columns,
    credit_scale=0.01,
    test_size=0.3,
    random_state=random_seed
)

# Normalize the data to ensure all features have a median of zero
X_train_normalized, X_test_normalized = normalize(X_train, X_test)

# Convert to tensors for use with PyTorch
X_train_tensor = torch.tensor(X_train_normalized.values, dtype=torch.float32,
↳device=device)
X_test_tensor = torch.tensor(X_test_normalized.values, dtype=torch.float32,
↳device=device)
y_train_tensor = torch.tensor(y_train.values, dtype=torch.float32,
↳device=device)
y_test_tensor = torch.tensor(y_test.values, dtype=torch.float32, device=device)

feature_names = [
    'Account Status',
    'Duration (Months)',
    'Credit History',
    'Credit Amount',
    'Savings/Bonds',
    'Employment Length',
    'Other Plans',
    'Foreign Worker',

```

```
'Unknown1',
'Unknown2'
]
```

1.3 3. Data Overview

Before building our models, we'll get an overview of our data.

```
[ ]: print("Shape of training data:", X_train.shape)
print("Shape of testing data:", X_test.shape)
print("\nClass distribution in training set:")
print(y_train.value_counts(normalize=True))
print("\nClass distribution in test set:")
print(y_test.value_counts(normalize=True))
```

```
Shape of training data: (700, 10)
Shape of testing data: (300, 10)
```

```
Class distribution in training set:
Class label
0    0.704286
1    0.295714
Name: proportion, dtype: float64
```

```
Class distribution in test set:
Class label
0    0.69
1    0.31
Name: proportion, dtype: float64
```

```
[ ]: feature_stats = feature_summary(X_train, categorical_threshold = 15)
print("Feature Statistics:")
print(feature_stats)
```

Feature Statistics:

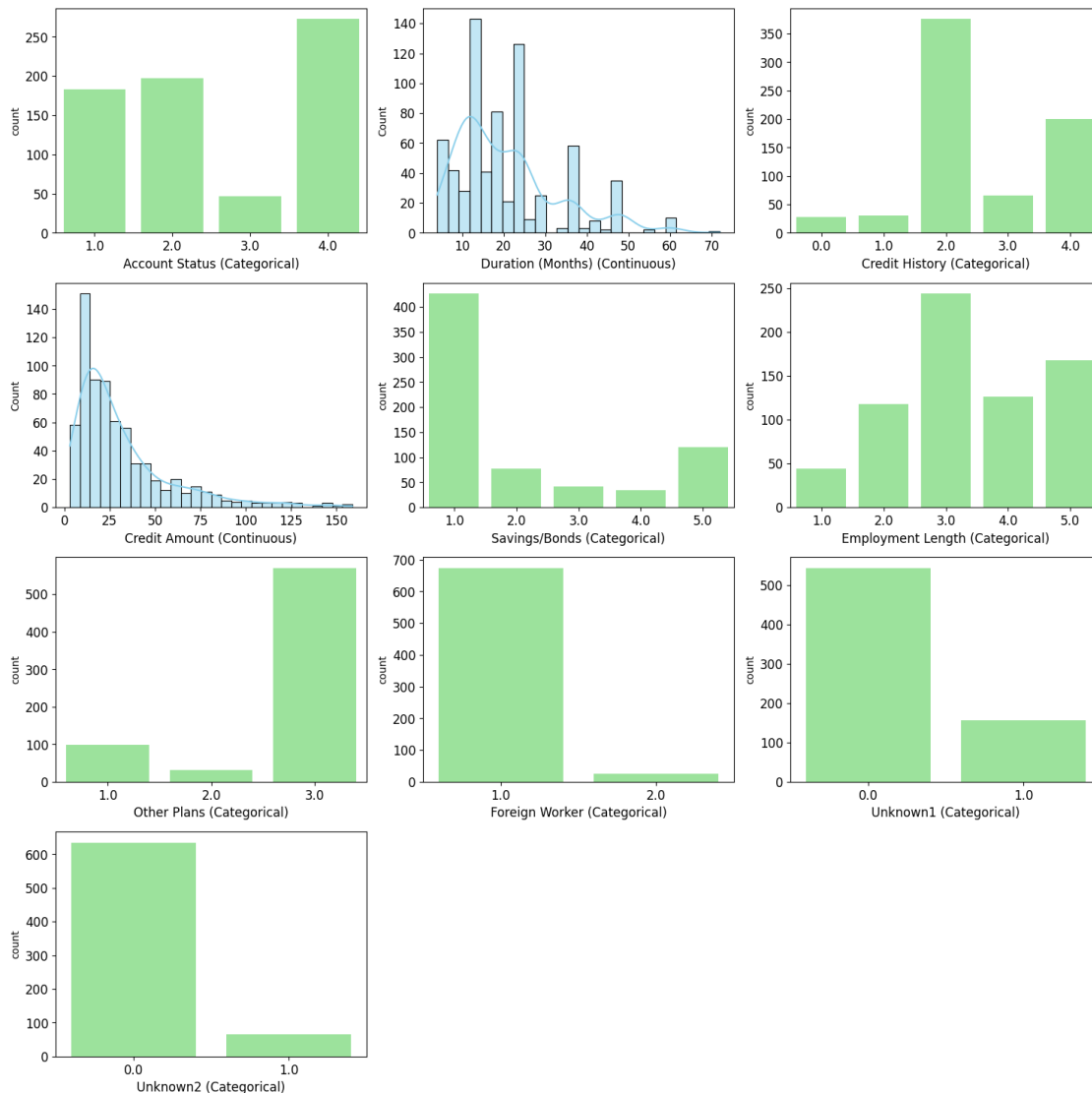
	Data Type	Non-Null Count	Null Count	\
Status of existing checking account	float64	700	0	
Duration in month	float64	700	0	
Credit history	float64	700	0	
Credit amount	float64	700	0	
Savings account/bonds	float64	700	0	
Period of present employment	float64	700	0	
Other installment plans	float64	700	0	
Foreign worker	float64	700	0	
Unknown1	float64	700	0	
unknown2	float64	700	0	

	Mean	Median	Std Dev	IQR	Min	\
Status of existing checking account	2.585714	2.0	1.244074	3.00	1.0	

Duration in month	20.652857	18.0	12.279683	12.00	4.0
Credit history	2.542857	2.0	1.070955	2.00	0.0
Credit amount	31.822857	23.0	27.278724	25.25	3.0
Savings account/bonds	2.061429	1.0	1.551389	2.00	1.0
Period of present employment	3.365714	3.0	1.195112	1.00	1.0
Other installment plans	2.674286	3.0	0.707347	0.00	1.0
Foreign worker	1.037143	1.0	0.189247	0.00	1.0
Unknown1	0.224286	0.0	0.417409	0.00	0.0
unknown2	0.092857	0.0	0.290440	0.00	0.0

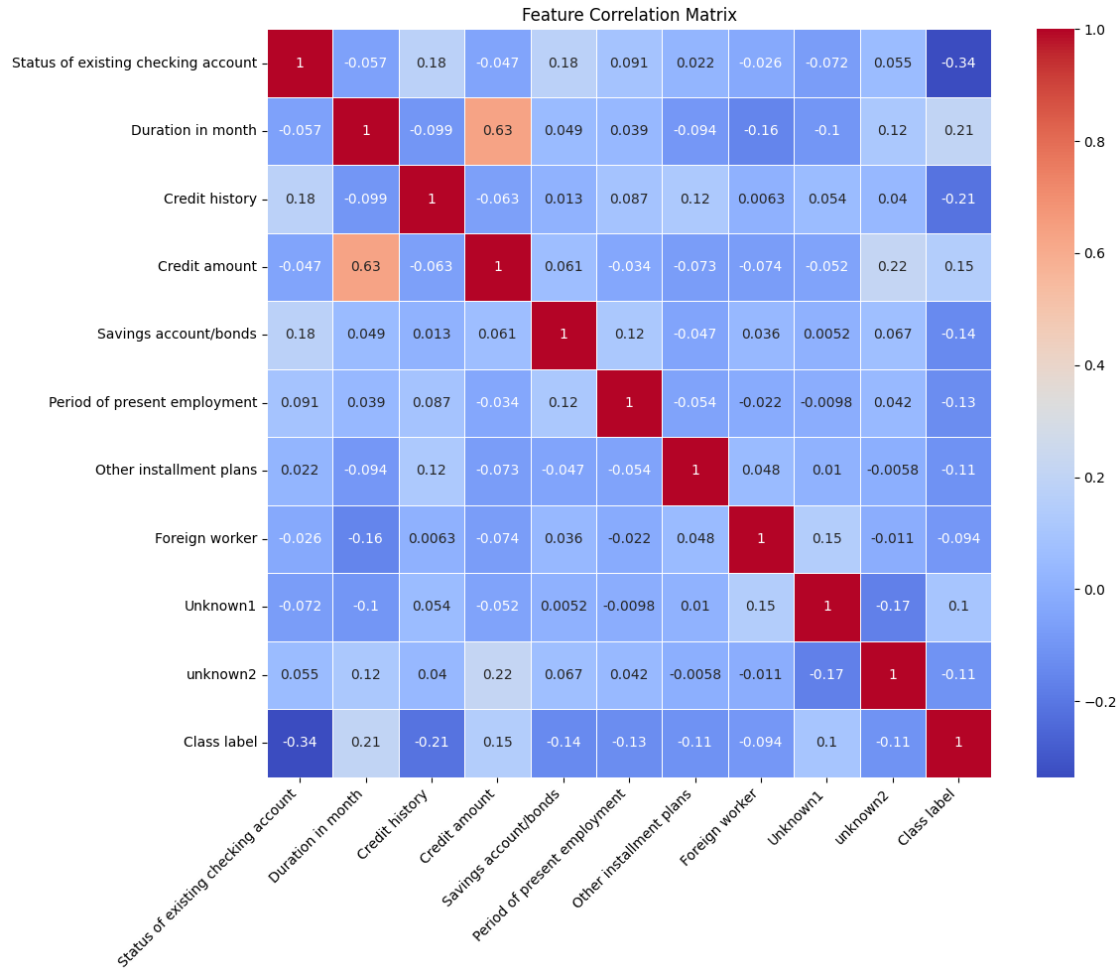
	Max	Unique Values	Is Categorical
Status of existing checking account	4.0	4	True
Duration in month	72.0	32	False
Credit history	4.0	5	True
Credit amount	159.0	112	False
Savings account/bonds	5.0	5	True
Period of present employment	5.0	5	True
Other installment plans	3.0	3	True
Foreign worker	2.0	2	True
Unknown1	1.0	2	True
unknown2	1.0	2	True

```
[ ]: plot_feature_histograms(X_train,
    ↪feature_stats,figsize=(15,15),feature_names=[*feature_names,'Class label'])
```



```
[ ]: <module 'matplotlib.pyplot' from
      'c:\\Users\\localuser\\PRISM\\prism_github\\venv_prism\\Lib\\site-
      packages\\matplotlib\\pyplot.py'>
```

```
[ ]: # Correlation matrix
correlation_matrix = pd.concat([X_train, y_train], axis=1).corr()
plt.figure(figsize=(12, 10))
sns.heatmap(correlation_matrix, annot=True, cmap='coolwarm', linewidths=0.5)
plt.title('Feature Correlation Matrix')
plt.xticks(rotation=45, ha='right')
plt.tight_layout()
plt.show()
```



1.4 4. Train Initial Black Box Model (MLP)

We start by training a black box model, in this case, a Multi-Layer Perceptron (MLP) with 10 nodes in the hidden layer. This model serves as our baseline and represents the type of complex, hard-to-interpret model that is often used in machine learning.

```
[ ]: mlp_hyperparameters = {
    'n_hidden': 10,
    'weight_decay': 0.0001,
    'lr': 0.001,
    'patience': 10,
    'tolerance': 0.00001,
    'batch_size': 32,
    'device': device,
    'seed': random_seed
}
```



```
blackbox_model = train_mlp_batched(X_train_normalized, y_train,  
↪X_test_normalized, y_test, **mlp_hyperparameters)
```

```
Epoch 0: Train loss 0.7623, Val loss 0.7541  
Epoch 1: Train loss 0.7124, Val loss 0.7045  
Epoch 2: Train loss 0.6719, Val loss 0.6644  
Epoch 3: Train loss 0.6382, Val loss 0.6327  
Epoch 4: Train loss 0.6130, Val loss 0.6061  
Epoch 5: Train loss 0.5907, Val loss 0.5855  
Epoch 6: Train loss 0.5744, Val loss 0.5692  
Epoch 7: Train loss 0.5612, Val loss 0.5558  
Epoch 8: Train loss 0.5499, Val loss 0.5451  
Epoch 9: Train loss 0.5415, Val loss 0.5359  
Epoch 10: Train loss 0.5342, Val loss 0.5282  
Epoch 11: Train loss 0.5271, Val loss 0.5220  
Epoch 12: Train loss 0.5226, Val loss 0.5164  
Epoch 13: Train loss 0.5185, Val loss 0.5119  
Epoch 14: Train loss 0.5133, Val loss 0.5073  
Epoch 15: Train loss 0.5108, Val loss 0.5036  
Epoch 16: Train loss 0.5062, Val loss 0.5005  
Epoch 17: Train loss 0.5043, Val loss 0.4977  
Epoch 18: Train loss 0.5014, Val loss 0.4953  
Epoch 19: Train loss 0.4994, Val loss 0.4930  
Epoch 20: Train loss 0.4980, Val loss 0.4913  
Epoch 21: Train loss 0.4956, Val loss 0.4894  
Epoch 22: Train loss 0.4945, Val loss 0.4878  
Epoch 23: Train loss 0.4941, Val loss 0.4865  
Epoch 24: Train loss 0.4918, Val loss 0.4852  
Epoch 25: Train loss 0.4900, Val loss 0.4844  
Epoch 26: Train loss 0.4884, Val loss 0.4831  
Epoch 27: Train loss 0.4884, Val loss 0.4827  
Epoch 28: Train loss 0.4879, Val loss 0.4814  
Epoch 29: Train loss 0.4862, Val loss 0.4808  
Epoch 30: Train loss 0.4855, Val loss 0.4802  
Epoch 31: Train loss 0.4849, Val loss 0.4796  
Epoch 32: Train loss 0.4841, Val loss 0.4792  
Epoch 33: Train loss 0.4836, Val loss 0.4789  
Epoch 34: Train loss 0.4833, Val loss 0.4785  
Epoch 35: Train loss 0.4827, Val loss 0.4783  
Epoch 36: Train loss 0.4821, Val loss 0.4778  
Epoch 37: Train loss 0.4826, Val loss 0.4776  
Epoch 38: Train loss 0.4816, Val loss 0.4773  
Epoch 39: Train loss 0.4815, Val loss 0.4772  
Epoch 40: Train loss 0.4812, Val loss 0.4768  
Epoch 41: Train loss 0.4801, Val loss 0.4769  
Epoch 42: Train loss 0.4800, Val loss 0.4767  
Epoch 43: Train loss 0.4798, Val loss 0.4763  
Epoch 44: Train loss 0.4796, Val loss 0.4764
```

Epoch 45: Train loss 0.4797, Val loss 0.4763
Epoch 46: Train loss 0.4794, Val loss 0.4762
Epoch 47: Train loss 0.4794, Val loss 0.4763
Epoch 48: Train loss 0.4787, Val loss 0.4762
Epoch 49: Train loss 0.4782, Val loss 0.4757
Epoch 50: Train loss 0.4777, Val loss 0.4762
Epoch 51: Train loss 0.4780, Val loss 0.4758
Epoch 52: Train loss 0.4769, Val loss 0.4758
Epoch 53: Train loss 0.4780, Val loss 0.4756
Epoch 54: Train loss 0.4779, Val loss 0.4757
Epoch 55: Train loss 0.4772, Val loss 0.4755
Epoch 56: Train loss 0.4765, Val loss 0.4754
Epoch 57: Train loss 0.4770, Val loss 0.4755
Epoch 58: Train loss 0.4761, Val loss 0.4755
Epoch 59: Train loss 0.4763, Val loss 0.4755
Epoch 60: Train loss 0.4760, Val loss 0.4754
Epoch 61: Train loss 0.4768, Val loss 0.4755
Epoch 62: Train loss 0.4751, Val loss 0.4752
Epoch 63: Train loss 0.4750, Val loss 0.4749
Epoch 64: Train loss 0.4771, Val loss 0.4754
Epoch 65: Train loss 0.4748, Val loss 0.4749
Epoch 66: Train loss 0.4750, Val loss 0.4748
Epoch 67: Train loss 0.4750, Val loss 0.4751
Epoch 68: Train loss 0.4748, Val loss 0.4748
Epoch 69: Train loss 0.4747, Val loss 0.4749
Epoch 70: Train loss 0.4737, Val loss 0.4750
Epoch 71: Train loss 0.4731, Val loss 0.4749
Epoch 72: Train loss 0.4738, Val loss 0.4750
Epoch 73: Train loss 0.4733, Val loss 0.4746
Epoch 74: Train loss 0.4734, Val loss 0.4747
Epoch 75: Train loss 0.4731, Val loss 0.4747
Epoch 76: Train loss 0.4730, Val loss 0.4748
Epoch 77: Train loss 0.4728, Val loss 0.4746
Epoch 78: Train loss 0.4724, Val loss 0.4748
Epoch 79: Train loss 0.4734, Val loss 0.4747
Epoch 80: Train loss 0.4729, Val loss 0.4748
Epoch 81: Train loss 0.4720, Val loss 0.4749
Epoch 82: Train loss 0.4717, Val loss 0.4746
Epoch 83: Train loss 0.4719, Val loss 0.4748
Epoch 84: Train loss 0.4708, Val loss 0.4744
Epoch 85: Train loss 0.4710, Val loss 0.4746
Epoch 86: Train loss 0.4717, Val loss 0.4743
Epoch 87: Train loss 0.4706, Val loss 0.4744
Epoch 88: Train loss 0.4703, Val loss 0.4745
Epoch 89: Train loss 0.4700, Val loss 0.4744
Epoch 90: Train loss 0.4708, Val loss 0.4744
Epoch 91: Train loss 0.4707, Val loss 0.4741
Epoch 92: Train loss 0.4695, Val loss 0.4745

```

Epoch 93: Train loss 0.4700, Val loss 0.4743
Epoch 94: Train loss 0.4693, Val loss 0.4742
Epoch 95: Train loss 0.4688, Val loss 0.4745
Epoch 96: Train loss 0.4691, Val loss 0.4743
Epoch 97: Train loss 0.4686, Val loss 0.4743
Epoch 98: Train loss 0.4690, Val loss 0.4741
Epoch 99: Train loss 0.4681, Val loss 0.4741
Epoch 100: Train loss 0.4682, Val loss 0.4742
Epoch 101: Train loss 0.4668, Val loss 0.4743
Epoch 102: Train loss 0.4678, Val loss 0.4744
Epoch 103: Train loss 0.4671, Val loss 0.4738
Epoch 104: Train loss 0.4688, Val loss 0.4743
Epoch 105: Train loss 0.4670, Val loss 0.4745
Epoch 106: Train loss 0.4673, Val loss 0.4741
Epoch 107: Train loss 0.4673, Val loss 0.4741
Epoch 108: Train loss 0.4662, Val loss 0.4741
Epoch 109: Train loss 0.4667, Val loss 0.4739
Epoch 110: Train loss 0.4658, Val loss 0.4740
Epoch 111: Train loss 0.4656, Val loss 0.4743
Epoch 112: Train loss 0.4664, Val loss 0.4741
Stopping early at epoch 103

```

```

[ ]: y_pred_train_blackbox = blackbox_model.predict(X_train_tensor)
     y_pred_test_blackbox = blackbox_model.predict(X_test_tensor)

     # Evaluate MLP
     results_mlp_train = evaluate_model_performance(y_train, y_pred_train_blackbox,
     ↪y_train, title="MLP - training data")
     results_mlp_test = evaluate_model_performance(y_test, y_pred_test_blackbox,
     ↪y_train, title="MLP - test data")

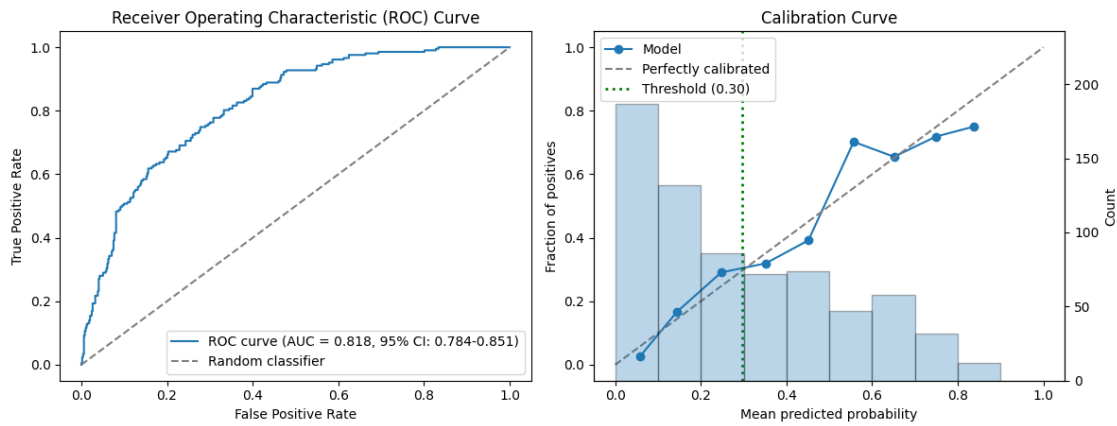
```

```

---- Model Performance Metrics MLP - training data ----
Threshold (prevalence) set to: 0.296
AUROC:          0.818 (95% CI: 0.784-0.851) (Area Under the Receiver Operating
Characteristic curve)
Accuracy:       0.723 (Proportion of correct predictions)
Precision:      0.522 (Proportion of true positives among positive predictions,
aka PPV)
Recall:         0.754 (Proportion of true positives among actual positives, aka
sensitivity)
F1 Score:       0.617 (Harmonic mean of precision and recall)
-----

```

MLP - training data



---- Model Performance Metrics MLP - test data ----

Threshold (prevalence) set to: 0.296

AUROC: 0.808 (95% CI: 0.756-0.856) (Area Under the Receiver Operating Characteristic curve)

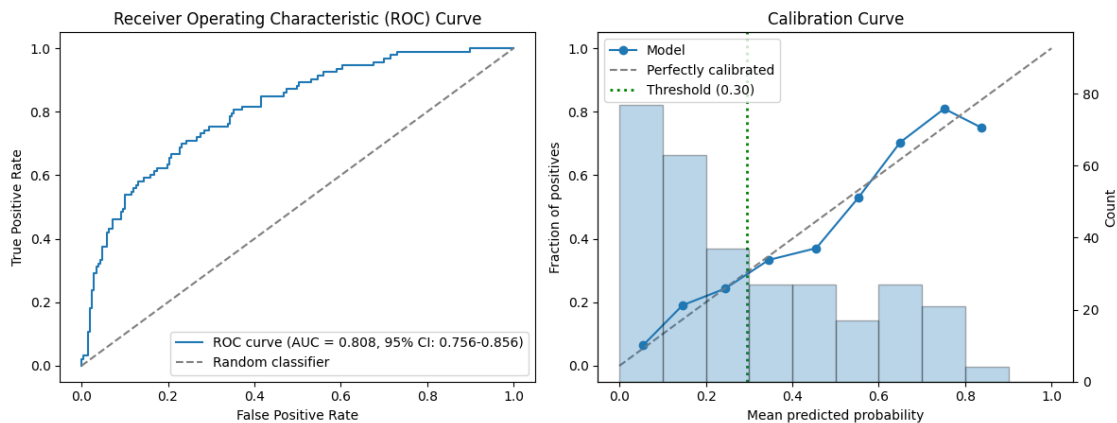
Accuracy: 0.723 (Proportion of correct predictions)

Precision: 0.540 (Proportion of true positives among positive predictions, aka PPV)

Recall: 0.720 (Proportion of true positives among actual positives, aka sensitivity)

F1 Score: 0.618 (Harmonic mean of precision and recall)

MLP - test data



1.5 5. Calculate Partial Responses

Now we calculate partial responses for our black box model. This is a key step in the PRiSM method. Partial responses help us understand how individual features or pairs of features contribute to the model's predictions, making the black box model more interpretable.

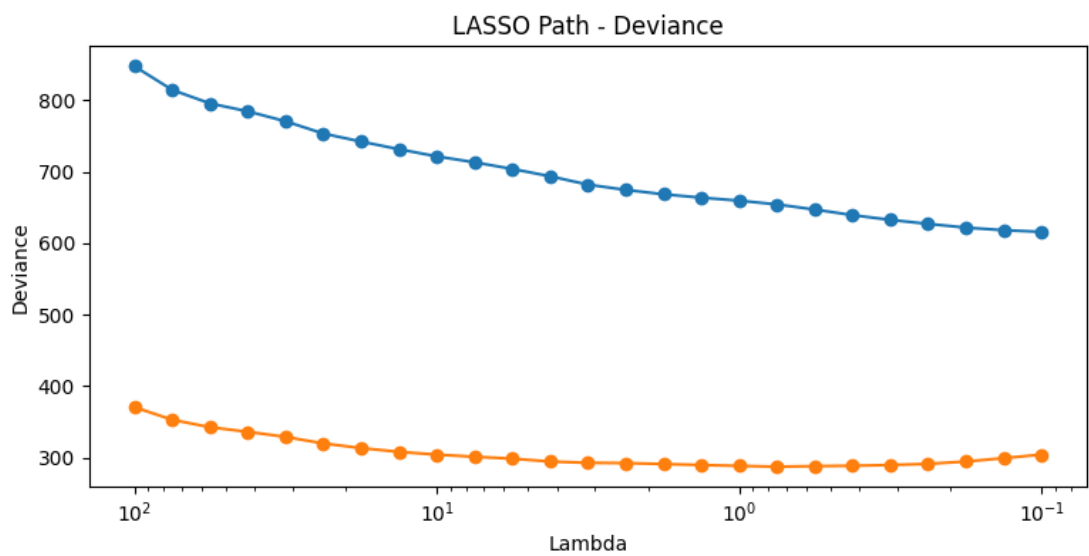
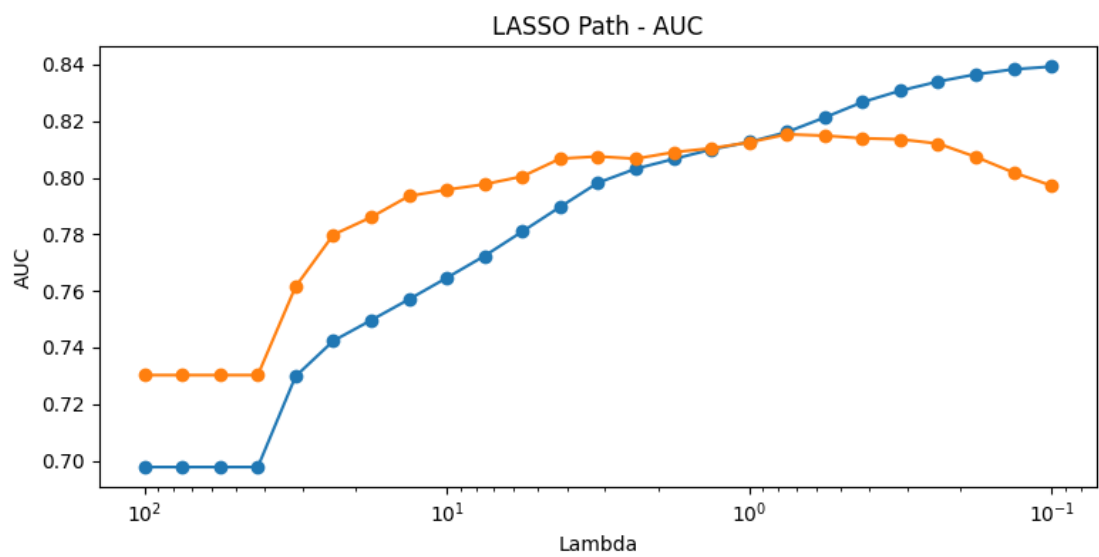
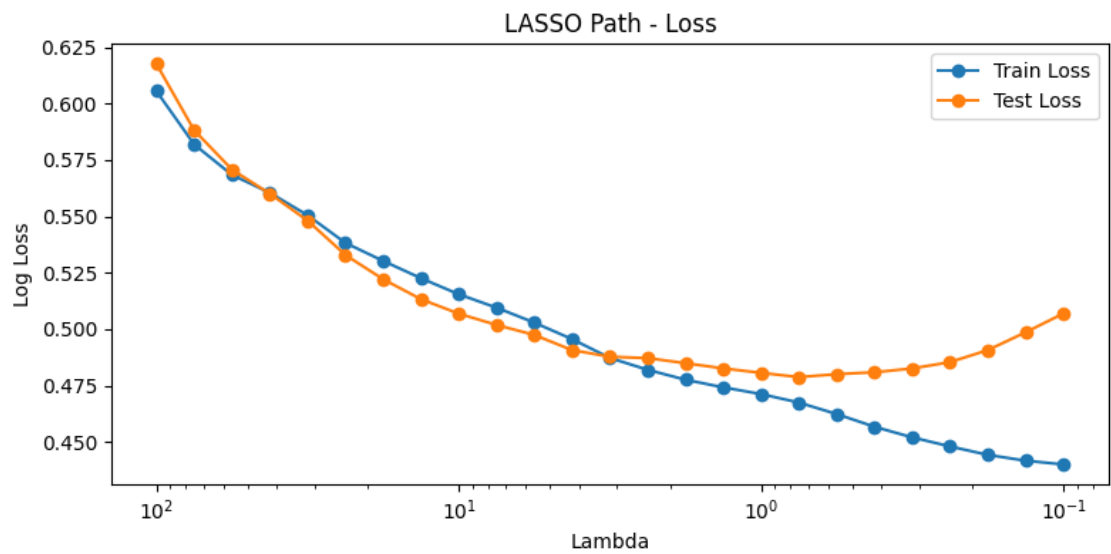
```
[ ]: partial_responses_params = {
    'x_train': X_train_tensor,
    'method': partial_response_method,
    'device': device,
    'batch_size': 512,
    'max_workers': max_workers
}

partial_responses_train = partial_responses(X_train_tensor, blackbox_model,
    ↪**partial_responses_params)
partial_responses_test = partial_responses(X_test_tensor, blackbox_model,
    ↪**partial_responses_params)
```

1.6 6. Perform LASSO on Partial Responses

We use LASSO (Least Absolute Shrinkage and Selection Operator) to select the most consequential partial responses. This step helps us identify which features or feature interactions contribute most to the model's predictions. A range of regularization strengths (lambda) are used, and for each one a logistic regression model with LASSO regularization is trained and evaluated. Based on the results, we can choose the most suitable lambda, balancing model performance against the total number of selected features.

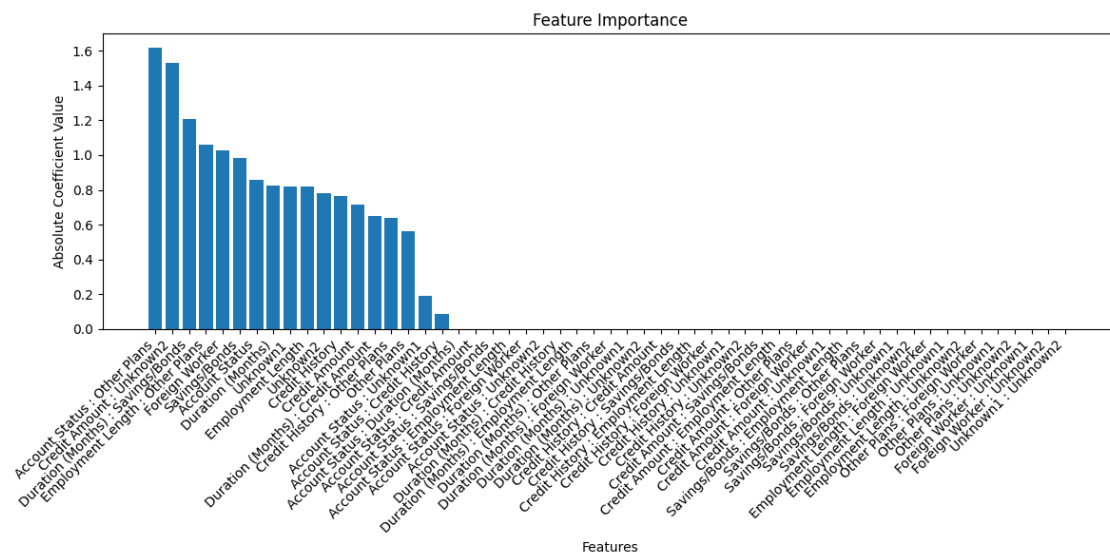
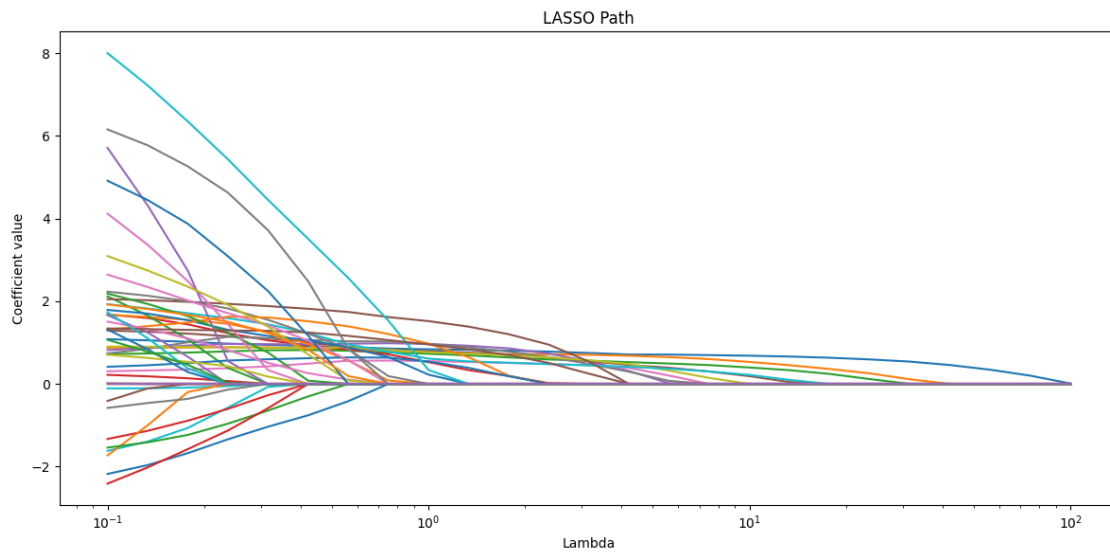
```
[ ]: lasso_results = lasso(
    partial_responses_train,
    partial_responses_test,
    y_train,
    y_test,
    feature_names=feature_names,
    nlambda=25,
    min_lambda=0.1,
    max_lambda=100,
    batch_size=2,
    seed=random_seed
)
```



<IPython.core.display.HTML object>

All fits converged successfully.

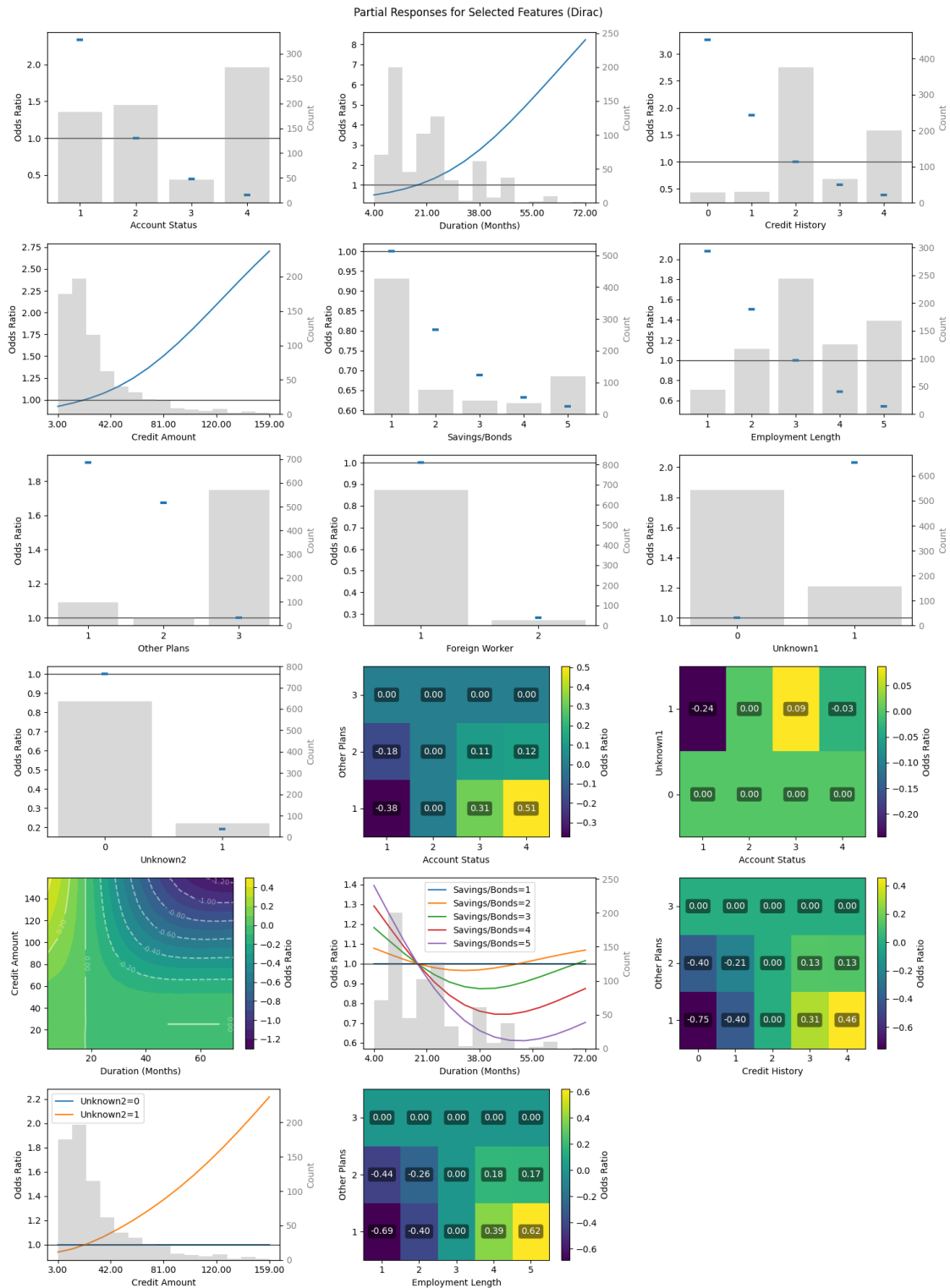
```
[ ]: lasso_results.plot_lambda_path()
lasso_results.select_lambda(17) # Adjust lambda selection as needed
lasso_results.plot_feature_importance()
```



1.7 7. View Selected Partial Responses and Generate Nomogram

First, we visualize the contribution of each partial response selected with LASSO to the target value.

```
[ ]: fig_blackbox = plot_partial_responses(  
    lasso_results,  
    X_train_tensor,  
    X_train.median().values,  
    X_train.std().values,  
    blackbox_model,  
    n_steps=15,  
    sd_scale=2,  
    method=partial_response_method,  
    device=device,  
    categorical_threshold=15,  
    subtract_univariate=True,  
    figsize=(15,20),  
    show_fig=True,  
    return_fig=True,  
    use_odds_ratio=True  
)
```

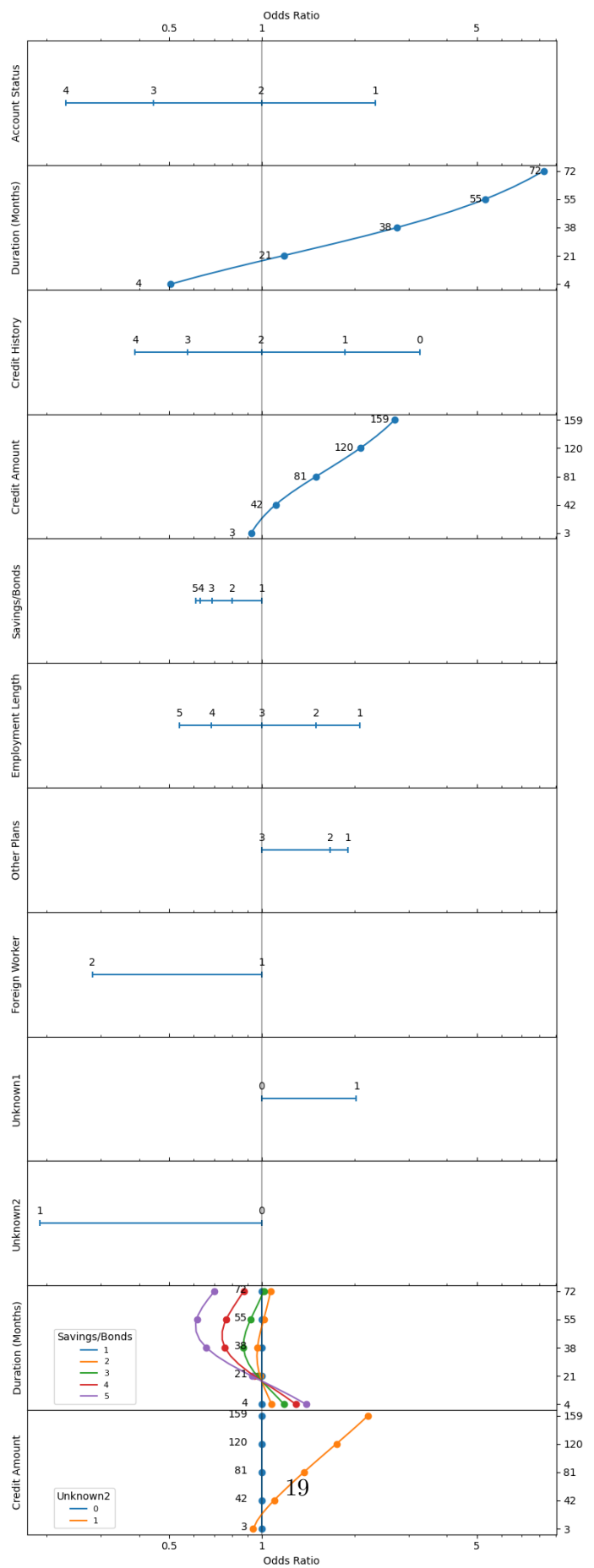



We can also visualize the contribution of each partial response using a nomogram. Nomograms are particularly useful in clinical settings as they provide an intuitive visual representation of a model's

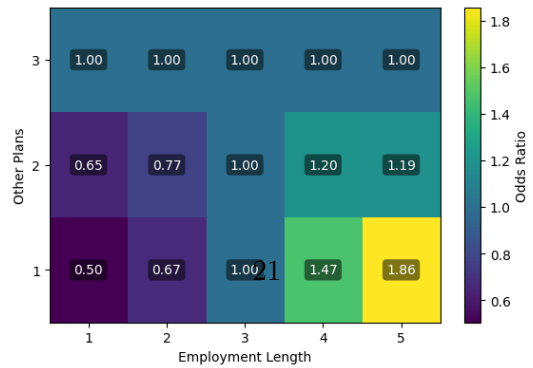
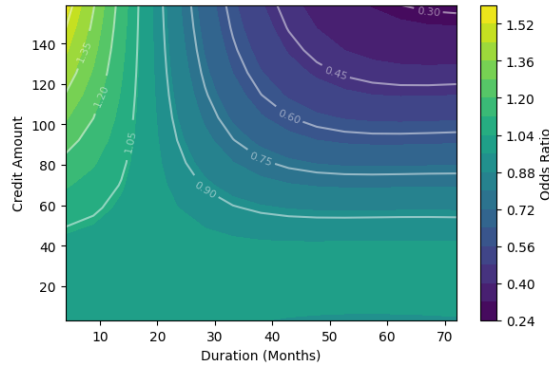
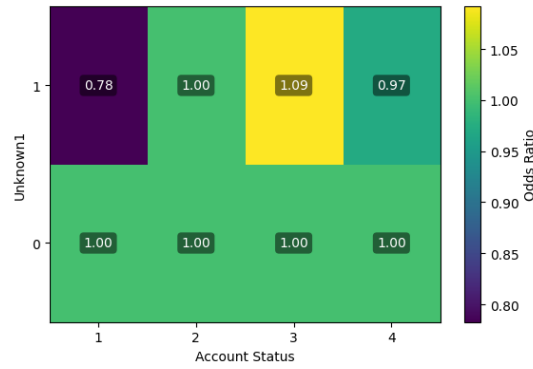
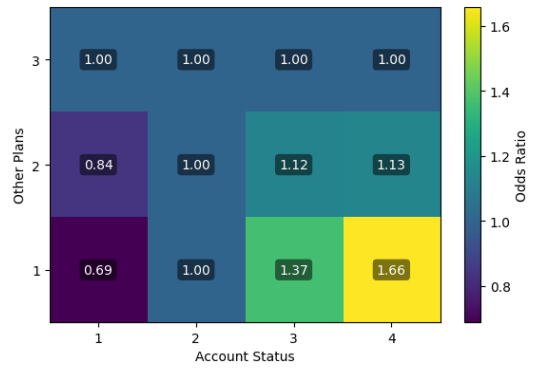
decision-making process.

```
[ ]: nomogram_params = {  
    'n_steps' : 15,  
    'sd_scale' : 2,  
    'method' : partial_response_method,  
    'device' : device,  
    'categorical_threshold' : 15,  
    'subtract_univariate' : True,  
    'return_fig' : True,  
    'use_odds_ratio' : True,  
    'save_csv' : SAVE_METRICS,  
    'csv_comment' : "MLP German CC data"  
}  
  
nomogram_results = nomogram(  
    lasso_results,  
    X_train_tensor,  
    X_train.median().values,  
    X_train.std().values,  
    blackbox_model,  
    **nomogram_params  
)  
nomogram_main_mlp = nomogram_results[6]  
nomogram_non_mixed_mlp = nomogram_results[7]
```

Nomogram of univariate and mixed bivariate partial responses (Dirac)



Nomogram of non-mixed bivariate partial responses (Dirac)



Univariate nomogram data saved to C:\Users\localuser\PRiSM\prism_github\models\nomogram_20240830_145532_univariate.csv

Bivariate nomogram data saved to C:\Users\localuser\PRiSM\prism_github\models\nomogram_20240830_145532_bivariate.csv

1.8 8. Train Partial Response Network (PRN)

Now we train a Partial Response Network based on the selected features. The PRN is a type of neural network that incorporates the interpretability of partial responses while maintaining the predictive power of the original black box model.

From the results of the lasso feature selection, we create an input mask for the PRN. The mask is a binary matrix that determines which input features connect to which hidden nodes (subnets) in the network. This enforces the network structure based on selected features.

The PRN uses a subnet architecture, where groups of hidden nodes are dedicated to specific features or feature interactions:

- Univariate subnets: Handle individual feature effects
- Bivariate subnets: Capture interactions between two features
- The number of nodes per subnet is controlled by the `subnet_nodes` parameter in `train_maskedmlp` (default 5)

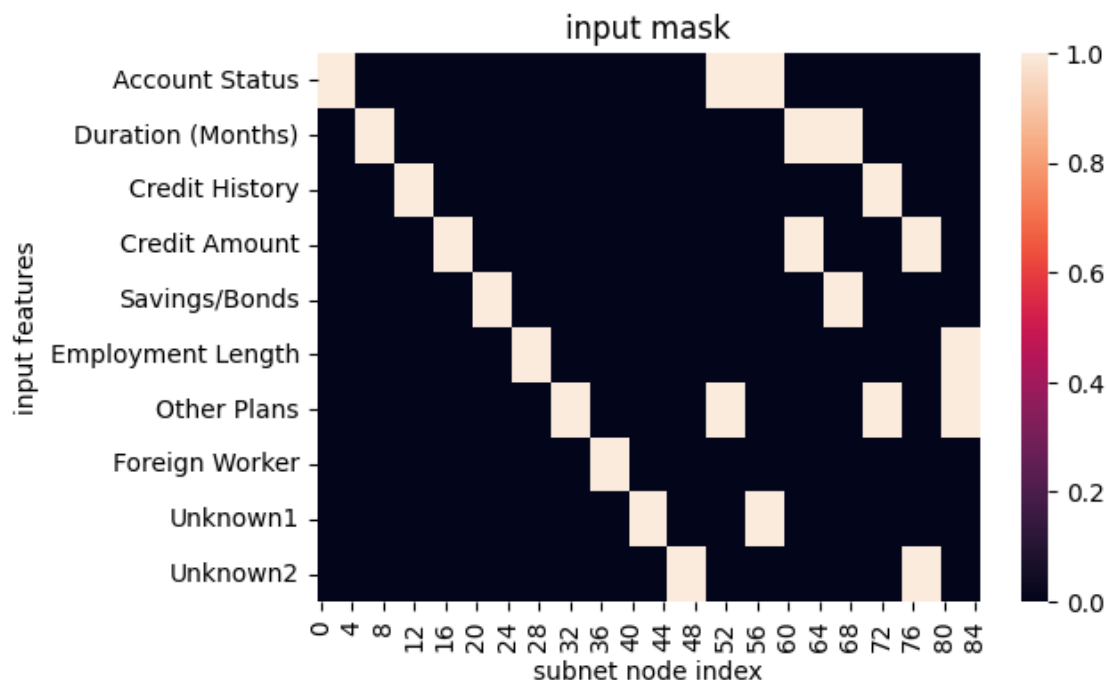
This gives us a resulting network structure as follows

- Input layer: Original features
- Hidden layer: Structured into subnets based on the mask
- Output layer: Single node for binary classification

The resulting PRN maintains interpretability because each subnet corresponds to a specific feature or interaction. Therefore, contribution of each feature/interaction can be analyzed separately, and the overall prediction is a sum of these interpretable components.

```
[ ]: mask, n_features = lasso_results.get_mask()
```

```
Selected features: ['Account Status', 'Duration (Months)', 'Credit History',  
'Credit Amount', 'Savings/Bonds', 'Employment Length', 'Other Plans', 'Foreign  
Worker', 'Unknown1', 'Unknown2', 'Account Status : Other Plans', 'Account Status  
: Unknown1', 'Duration (Months) : Credit Amount', 'Duration (Months) :  
Savings/Bonds', 'Credit History : Other Plans', 'Credit Amount : Unknown2',  
'Employment Length : Other Plans']
```



```
[ ]: prn_hyperparameters = {
    'n_hidden': n_features,
    'mask': mask,
    'subnet_nodes': 5,
    'iter': 10000,
    'lr': 0.001,
    'weight_decay': 0.00001,
    'tolerance': 0.00001,
    'patience': 100,
    'device': device,
    'seed': random_seed
}

partial_response_network = train_maskedmlp(X_train_normalized, y_train,
↪X_test_normalized, y_test, **prn_hyperparameters)
```

```
Epoch 0, Training loss 0.7236233949661255, Test loss 0.7323100566864014
Epoch 1, Training loss 0.7328987121582031, Test loss 0.7280257344245911
Epoch 2, Training loss 0.7286074161529541, Test loss 0.7237021327018738
Epoch 3, Training loss 0.7242798209190369, Test loss 0.7193962335586548
Epoch 4, Training loss 0.7199699878692627, Test loss 0.7151332497596741
Epoch 5, Training loss 0.7157022953033447, Test loss 0.7109275460243225
Epoch 6, Training loss 0.7114904522895813, Test loss 0.7067879438400269
Epoch 7, Training loss 0.7073429822921753, Test loss 0.7027204036712646
Epoch 8, Training loss 0.7032657265663147, Test loss 0.6987292170524597
```

Epoch 9, Training loss 0.6992627382278442, Test loss 0.6948171854019165
Epoch 10, Training loss 0.6953367590904236, Test loss 0.6909867525100708
Epoch 11, Training loss 0.6914900541305542, Test loss 0.6872389316558838
Epoch 12, Training loss 0.6877239346504211, Test loss 0.6835748553276062
Epoch 13, Training loss 0.6840395331382751, Test loss 0.6799950003623962
Epoch 14, Training loss 0.6804371476173401, Test loss 0.676499605178833
Epoch 15, Training loss 0.6769170761108398, Test loss 0.6730881929397583
Epoch 16, Training loss 0.6734791398048401, Test loss 0.6697606444358826
Epoch 17, Training loss 0.6701231002807617, Test loss 0.6665160655975342
Epoch 18, Training loss 0.6668483018875122, Test loss 0.6633538007736206
Epoch 19, Training loss 0.663654088973999, Test loss 0.6602726578712463
Epoch 20, Training loss 0.6605395674705505, Test loss 0.6572718024253845
Epoch 21, Training loss 0.6575037240982056, Test loss 0.6543497443199158
Epoch 22, Training loss 0.6545454859733582, Test loss 0.6515051126480103
Epoch 23, Training loss 0.6516635417938232, Test loss 0.6487366557121277
Epoch 24, Training loss 0.6488566994667053, Test loss 0.6460426449775696
Epoch 25, Training loss 0.6461234092712402, Test loss 0.6434214115142822
Epoch 26, Training loss 0.6434623599052429, Test loss 0.6408714652061462
Epoch 27, Training loss 0.6408720016479492, Test loss 0.6383908987045288
Epoch 28, Training loss 0.6383507251739502, Test loss 0.6359779834747314
Epoch 29, Training loss 0.6358969807624817, Test loss 0.6336308717727661
Epoch 30, Training loss 0.6335089206695557, Test loss 0.6313477158546448
Epoch 31, Training loss 0.631184995174408, Test loss 0.6291266083717346
Epoch 32, Training loss 0.6289235353469849, Test loss 0.6269655227661133
Epoch 33, Training loss 0.6267225742340088, Test loss 0.624862790107727
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Epoch 41, Training loss 0.6110861897468567, Test loss 0.6099059581756592
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Epoch 43, Training loss 0.6076613068580627, Test loss 0.6066176891326904
Epoch 44, Training loss 0.6060124635696411, Test loss 0.6050317883491516
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Epoch 62, Training loss 0.5819790363311768, Test loss 0.5815041661262512
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Epoch 231, Training loss 0.5009151697158813, Test loss 0.49563828110694885
Epoch 232, Training loss 0.5006925463676453, Test loss 0.4954182803630829
Epoch 233, Training loss 0.500471830368042, Test loss 0.495200514793396
Epoch 234, Training loss 0.5002530813217163, Test loss 0.4949847459793091
Epoch 235, Training loss 0.5000361800193787, Test loss 0.49477121233940125
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Epoch 1335, Training loss 0.4574747085571289, Test loss 0.4745841324329376
Epoch 1336, Training loss 0.45745593309402466, Test loss 0.47458475828170776
Epoch 1337, Training loss 0.4574371874332428, Test loss 0.4745852053165436
Epoch 1338, Training loss 0.4574185311794281, Test loss 0.4745858907699585
Epoch 1339, Training loss 0.457399845123291, Test loss 0.4745864272117615
Epoch 1340, Training loss 0.4573812484741211, Test loss 0.4745870530605316
Epoch 1341, Training loss 0.45736268162727356, Test loss 0.4745877683162689
Epoch 1342, Training loss 0.45734408497810364, Test loss 0.4745884835720062
Epoch 1343, Training loss 0.4573255777359009, Test loss 0.47458913922309875
Epoch 1344, Training loss 0.4573070704936981, Test loss 0.47458988428115845
Epoch 1345, Training loss 0.45728859305381775, Test loss 0.4745907187461853
Epoch 1346, Training loss 0.45727014541625977, Test loss 0.47459152340888977
Epoch 1347, Training loss 0.45725181698799133, Test loss 0.474592387676239
Epoch 1348, Training loss 0.45723336935043335, Test loss 0.4745931923389435
Epoch 1349, Training loss 0.4572150409221649, Test loss 0.47459420561790466
Epoch 1350, Training loss 0.45719674229621887, Test loss 0.4745950698852539
Epoch 1351, Training loss 0.4571784436702728, Test loss 0.4745960533618927
Epoch 1352, Training loss 0.45716017484664917, Test loss 0.4745970070362091

Epoch 1353, Training loss 0.4571419060230255, Test loss 0.47459807991981506
Epoch 1354, Training loss 0.4571237564086914, Test loss 0.47459909319877625
Epoch 1355, Training loss 0.45710548758506775, Test loss 0.474600225687027
Epoch 1356, Training loss 0.4570873975753784, Test loss 0.47460129857063293
Epoch 1357, Training loss 0.4570692777633667, Test loss 0.47460246086120605
Epoch 1358, Training loss 0.45705121755599976, Test loss 0.4746036231517792
Epoch 1359, Training loss 0.4570331275463104, Test loss 0.47460484504699707
Epoch 1360, Training loss 0.45701512694358826, Test loss 0.47460606694221497
Epoch 1361, Training loss 0.4569970667362213, Test loss 0.47460734844207764
Epoch 1362, Training loss 0.45697909593582153, Test loss 0.47460857033729553
Epoch 1363, Training loss 0.45696115493774414, Test loss 0.4746098816394806
Epoch 1364, Training loss 0.45694321393966675, Test loss 0.47461119294166565
Epoch 1365, Training loss 0.45692530274391174, Test loss 0.47461262345314026
Epoch 1366, Training loss 0.4569075107574463, Test loss 0.4746139943599701
Epoch 1367, Training loss 0.45688965916633606, Test loss 0.4746153652667999
Epoch 1368, Training loss 0.4568718671798706, Test loss 0.4746169149875641
Epoch 1369, Training loss 0.4568540155887604, Test loss 0.4746183156967163
Epoch 1370, Training loss 0.4568363428115845, Test loss 0.47461989521980286
Epoch 1371, Training loss 0.4568185806274414, Test loss 0.47462138533592224
Epoch 1372, Training loss 0.4568008780479431, Test loss 0.4746229946613312
Epoch 1373, Training loss 0.4567831754684448, Test loss 0.47462454438209534
Epoch 1374, Training loss 0.4567655622959137, Test loss 0.4746261537075043
Epoch 1375, Training loss 0.45674794912338257, Test loss 0.4746277928352356
Epoch 1376, Training loss 0.4567303955554962, Test loss 0.4746294617652893
Epoch 1377, Training loss 0.4567128121852875, Test loss 0.47463124990463257
Epoch 1378, Training loss 0.4566952884197235, Test loss 0.47463297843933105
Epoch 1379, Training loss 0.45667776465415955, Test loss 0.47463470697402954
Epoch 1380, Training loss 0.4566602408885956, Test loss 0.4746364951133728
Epoch 1381, Training loss 0.4566428065299988, Test loss 0.47463831305503845
Epoch 1382, Training loss 0.456625372171402, Test loss 0.4746401011943817
Epoch 1383, Training loss 0.45660796761512756, Test loss 0.47464197874069214
Epoch 1384, Training loss 0.4565906226634979, Test loss 0.47464385628700256
Epoch 1385, Training loss 0.4565732181072235, Test loss 0.47464579343795776
Epoch 1386, Training loss 0.45655587315559387, Test loss 0.47464773058891296
Epoch 1387, Training loss 0.4565386176109314, Test loss 0.47464969754219055
Epoch 1388, Training loss 0.45652133226394653, Test loss 0.4746516942977905
Epoch 1389, Training loss 0.45650407671928406, Test loss 0.4746536910533905
Epoch 1390, Training loss 0.45648685097694397, Test loss 0.47465574741363525
Epoch 1391, Training loss 0.4564696252346039, Test loss 0.4746578335762024
Epoch 1392, Training loss 0.45645245909690857, Test loss 0.47465991973876953
Epoch 1393, Training loss 0.45643532276153564, Test loss 0.47466206550598145
Epoch 1394, Training loss 0.4564181864261627, Test loss 0.47466421127319336
Epoch 1395, Training loss 0.4564010500907898, Test loss 0.47466644644737244
Epoch 1396, Training loss 0.45638397336006165, Test loss 0.47466859221458435
Early stopping! Epoch 1396
Best model at epoch 1296

```
[ ]: # Evaluate PRN on training data
y_pred_train_prn = partial_response_network.predict(X_train_tensor,
↪device=device)

results_prn_train = evaluate_model_performance(y_train, y_pred_train_prn,
↪y_train, title="PRN - train data")

# Evaluate PRN on test data and compare to MLP (blackbox)
y_pred_test_prn = partial_response_network.predict(X_test_tensor, device=device)

results_prn_mlp_test = compare_model_performance(y_test, y_pred_test_blackbox,
↪y_pred_test_prn, y_train=y_train, model_names=("MLP", "PRN"), title="MLP-PRN
↪test comparison")
```

---- Model Performance Metrics PRN - train data ----

Threshold (prevalence) set to: 0.296

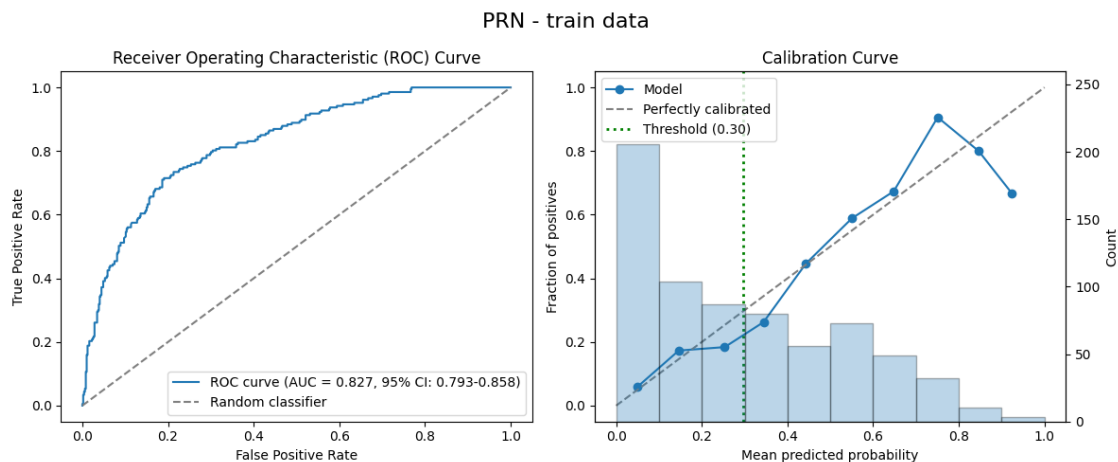
AUROC: 0.827 (95% CI: 0.793-0.858) (Area Under the Receiver Operating Characteristic curve)

Accuracy: 0.731 (Proportion of correct predictions)

Precision: 0.531 (Proportion of true positives among positive predictions, aka PPV)

Recall: 0.787 (Proportion of true positives among actual positives, aka sensitivity)

F1 Score: 0.634 (Harmonic mean of precision and recall)



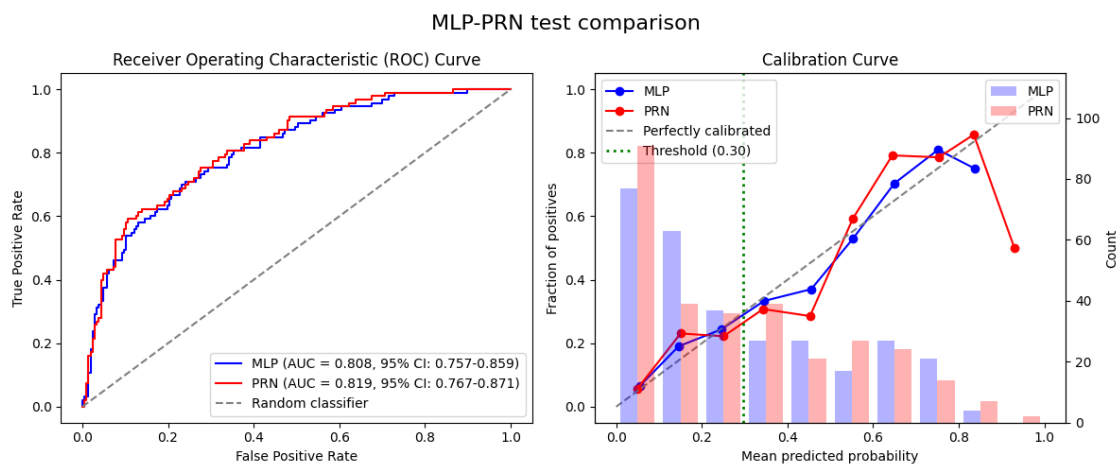
----- Model Performance Comparison -----

Threshold (prevalence) set to: 0.296

MLP:

AUROC: 0.808 (95% CI: 0.757-0.859)
 Accuracy: 0.723
 Precision: 0.540
 Recall: 0.720
 F1 Score: 0.618

PRN:
 AUROC: 0.819 (95% CI: 0.767-0.871)
 Accuracy: 0.720
 Precision: 0.533
 Recall: 0.774
 F1 Score: 0.632



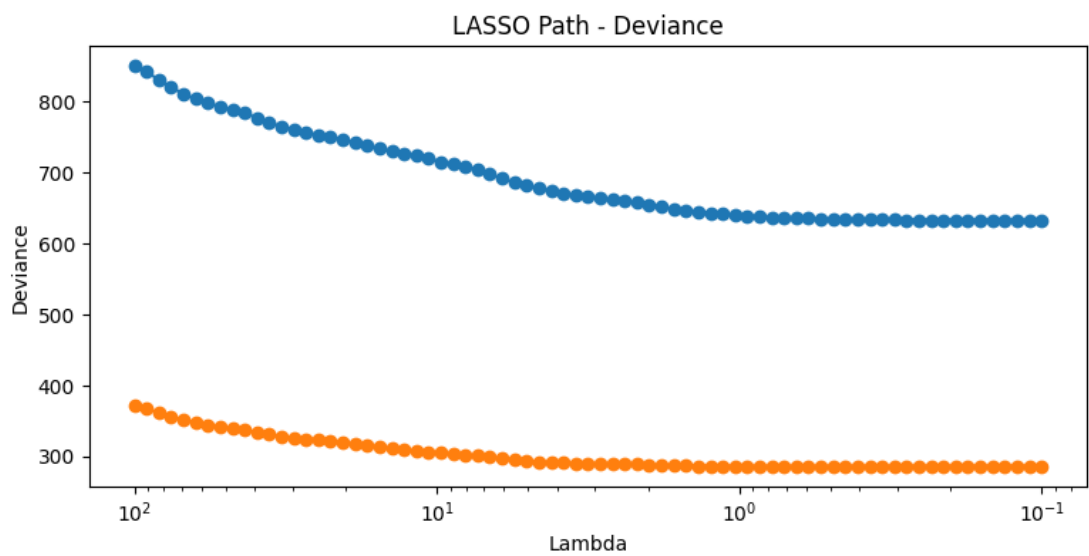
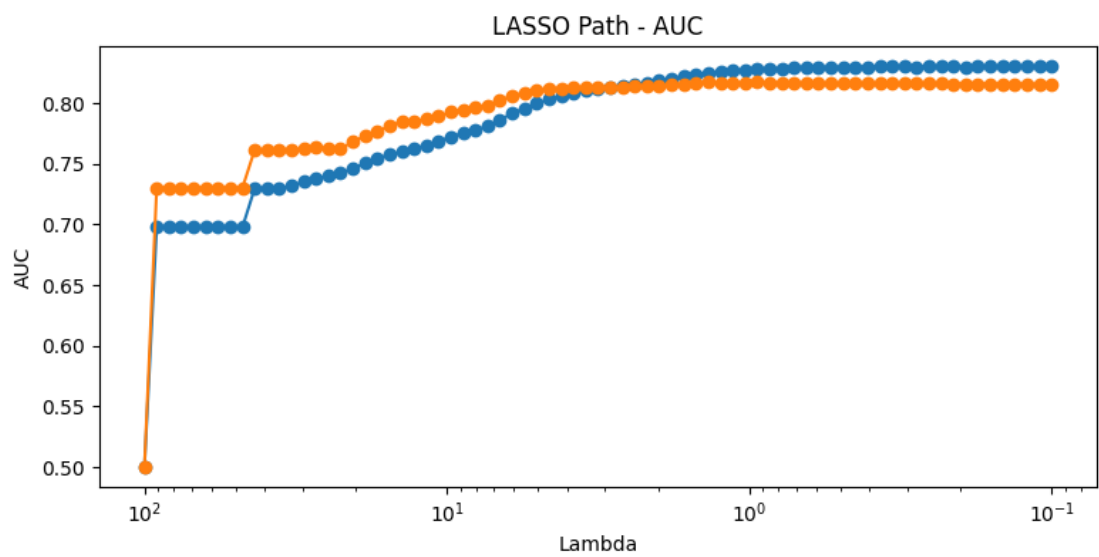
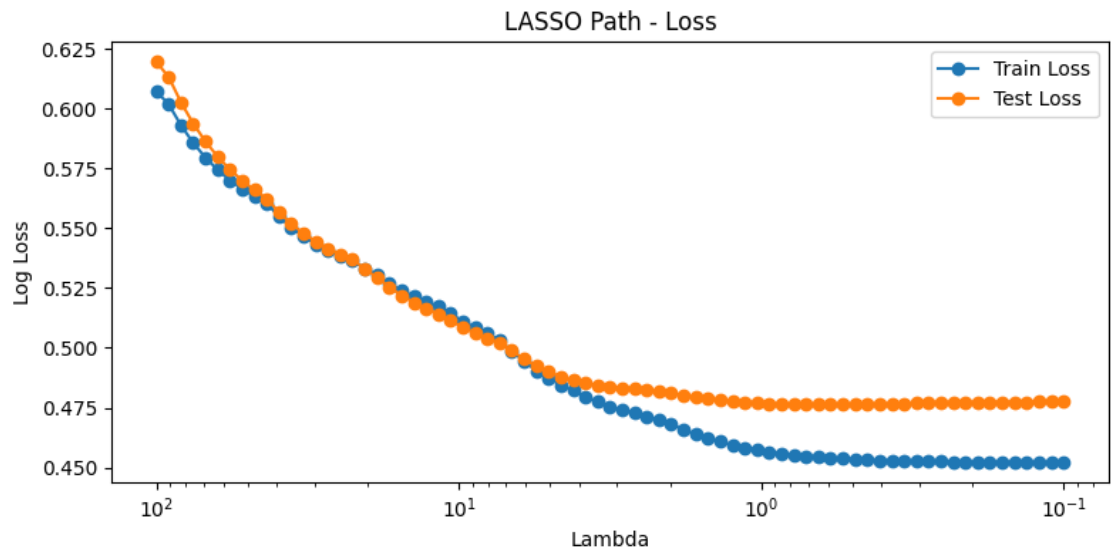
1.9 9. LASSO on PRN

We perform LASSO again, this time on the Partial Response Network to isolate the most important features and interactions.

```
[ ]: partial_responses_params_prn = {
    'x_train': X_train_tensor,
    'method': partial_response_method,
    'device': device,
    'batch_size': 256,
    'max_workers': max_workers
}

partial_responses_train_prn = partial_responses(X_train_tensor,
    ↪ partial_response_network, **partial_responses_params_prn)
partial_responses_test_prn = partial_responses(X_test_tensor,
    ↪ partial_response_network, **partial_responses_params_prn)
```

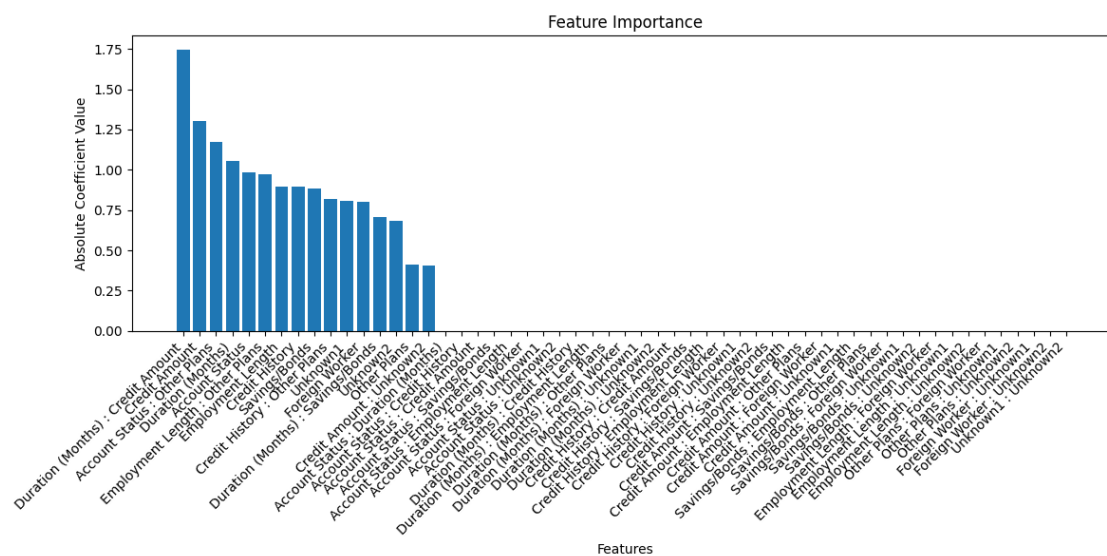
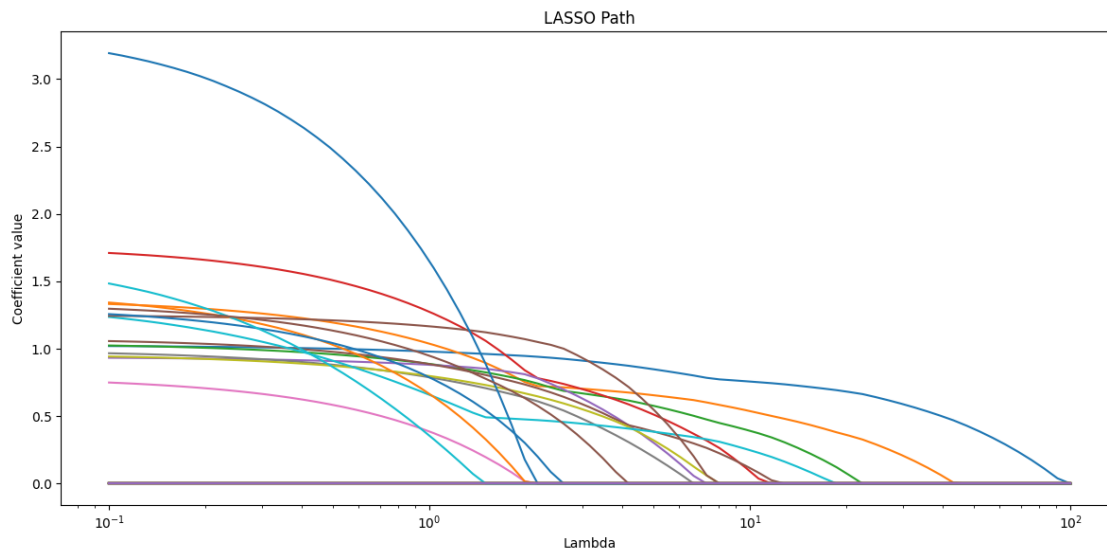
```
[ ]: lasso_results_prn = lasso(  
    partial_responses_train_prn,  
    partial_responses_test_prn,  
    y_train,  
    y_test,  
    feature_names=feature_names,  
    nlambdas=75,  
    min_lambda=0.1,  
    max_lambda=100,  
    batch_size=2,  
    seed=random_seed  
)
```



<IPython.core.display.HTML object>

All fits converged successfully.

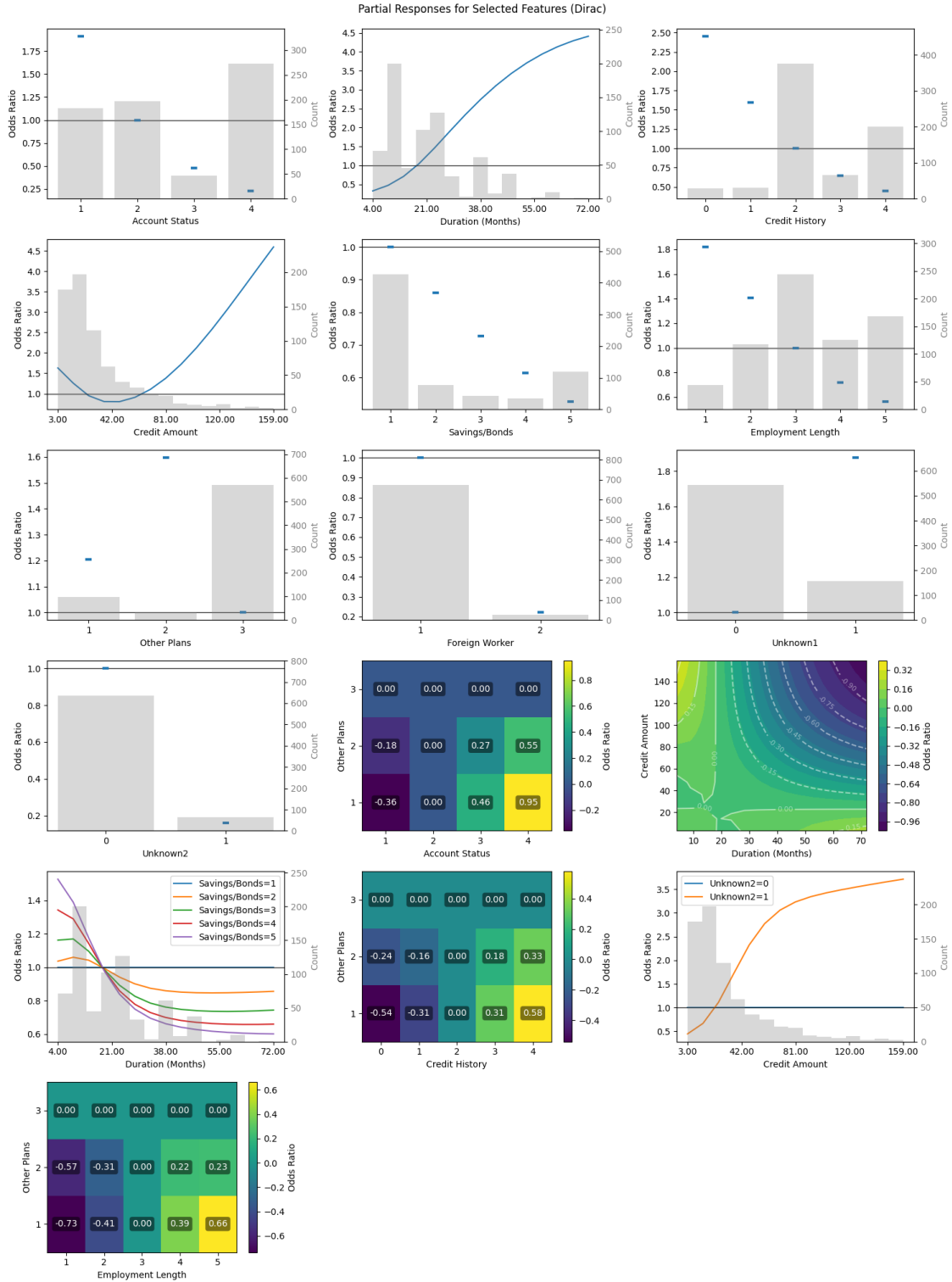
```
[ ]: lasso_results_prn.plot_lambda_path()
lasso_results_prn.select_lambda(50) # Adjust lambda selection as needed
lasso_results_prn.plot_feature_importance()
```



1.10 10. Generate Final Nomogram

Finally, we generate a nomogram for our PRN, based on the features selected with LASSO. The nomogram provides a visual tool that can be used to understand the model's decisions.

```
[ ]: fig_prn = plot_partial_responses(  
    lasso_results_prn,  
    X_train_tensor,  
    X_train.median().values,  
    X_train.std().values,  
    partial_response_network,  
    n_steps=15,  
    sd_scale=2,  
    method=partial_response_method,  
    device=device,  
    categorical_threshold=15,  
    subtract_univariate=True,  
    figsize=(15,20),  
    show_fig=True,  
    return_fig=True,  
    use_odds_ratio=True  
)
```



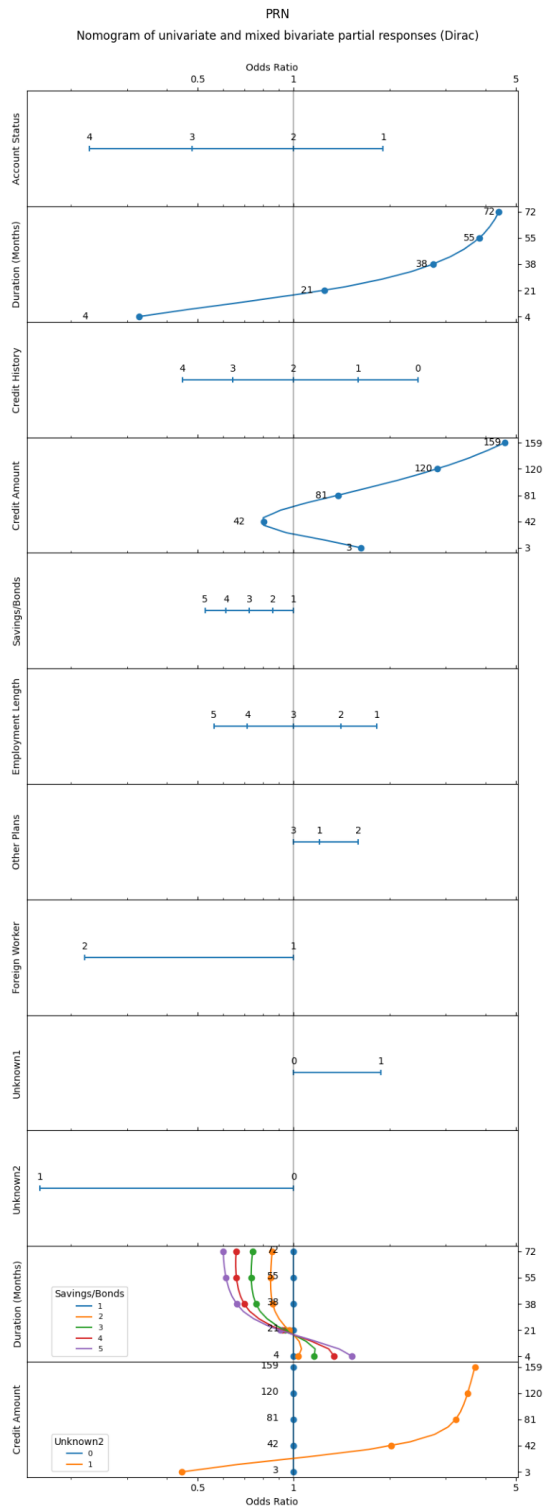
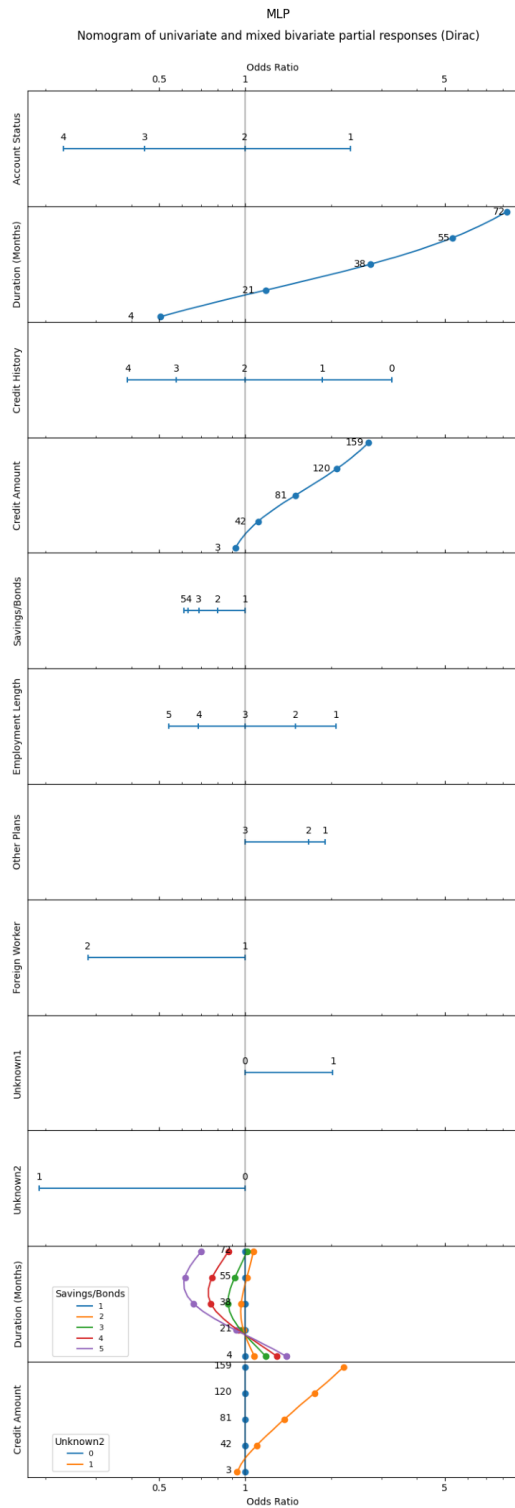
```
[ ]: %%capture
# ^ Supress nomogram plot output, will display side-by-side instead

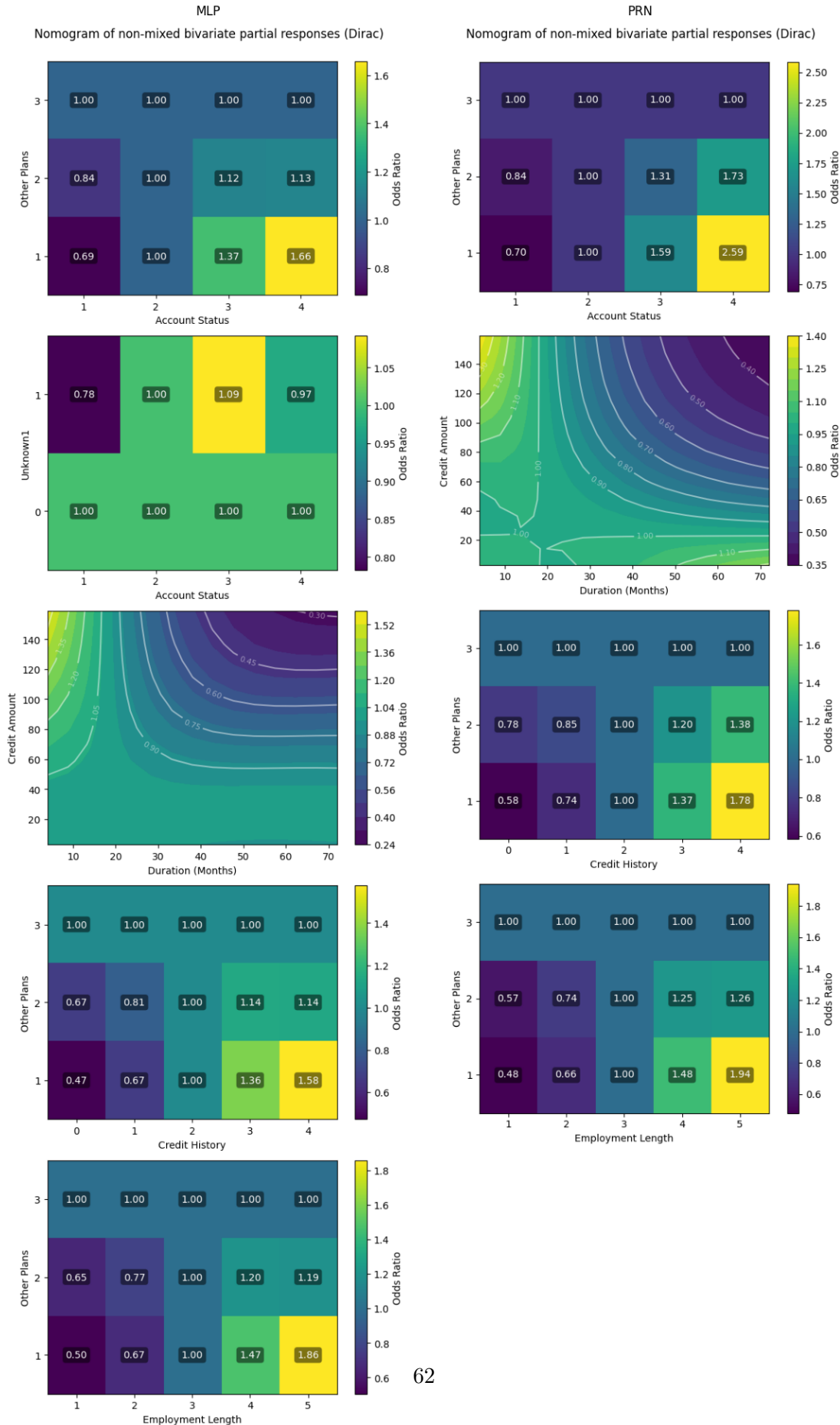
nomogram_params_prn = {
    'n_steps' : 15,
    'sd_scale' : 2,
    'method' : partial_response_method,
    'device' : device,
    'categorical_threshold' : 15,
    'subtract_univariate' : True,
    'show_fig' : False,
    'return_fig' : True,
    'use_odds_ratio' : True,
    'save_csv' : SAVE_METRICS,
    'csv_comment' : "PRN German CC data"
}

nomogram_results_prn = nomogram(
    lasso_results_prn,
    X_train_tensor,
    X_train.median().values,
    X_train.std().values,
    partial_response_network,
    **nomogram_params_prn
)

nomogram_main_prn = nomogram_results_prn[6]
nomogram_non_mixed_prn = nomogram_results_prn[7]

[ ]: display_nomograms_side_by_side(nomogram_main_mlp, nomogram_non_mixed_mlp,
                                     nomogram_main_prn, nomogram_non_mixed_prn)
```





1.11 11. Evaluate PRN-LASSO Model

Finally, we evaluate the simplified model (multivariate logistic regression with LASSO regularization) trained on the partial responses of the PRN, based on the selected lambda.

This notebook demonstrates the PRiSM method, transforming a black box model into an interpretable one without sacrificing predictive performance. This approach is particularly valuable when understanding the model's decision-making process is crucial for building trust and gaining insights.

The final PRN-LASSO model provides both high predictive accuracy and interpretability, allowing us to understand how different features contribute to the predictions.

```
[ ]: prn_lasso = lasso_results_prn.get_selected_model()
```

Logistic regression model for Lambda index 50 (0.9397) selected.

```
[ ]: # Evaluate PRN-LASSO on training data
y_pred_train_prn_lasso = prn_lasso.predict_proba(partial_responses_train_prn)[: , 1]

results_prn_lasso_train = evaluate_model_performance(y_train,
    ↪ y_pred_train_prn_lasso, y_train, title="PRN LASSO - train data")

# Evaluate PRN-LASSO on test data and compare to PRN
y_pred_test_prn_lasso = prn_lasso.predict_proba(partial_responses_test_prn)[: ,
    ↪ 1]

results_prn_prn_lasso_test = compare_model_performance(y_test, y_pred_test_prn,
    ↪ y_pred_test_prn_lasso, y_train=y_train, model_names=("PRN", "PRN LASSO"),
    ↪ title="PRN - PRN LASSO test comparison")

# Compare PRN-LASSO and original MLP (blackbox) model
results_prn_mlp_test = compare_model_performance(y_test, y_pred_test_blackbox,
    ↪ y_pred_test_prn_lasso, y_train=y_train, model_names=("MLP", "PRN LASSO"),
    ↪ title="MLP - PRN LASSO test comparison")
```

---- Model Performance Metrics PRN LASSO - train data ----

Threshold (prevalence) set to: 0.296

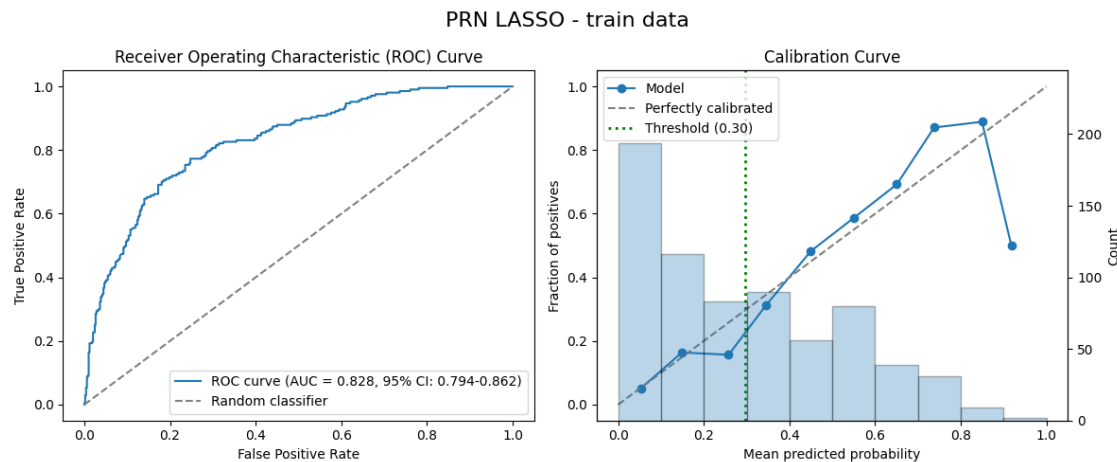
AUROC: 0.828 (95% CI: 0.794-0.862) (Area Under the Receiver Operating Characteristic curve)

Accuracy: 0.734 (Proportion of correct predictions)

Precision: 0.534 (Proportion of true positives among positive predictions, aka PPV)

Recall: 0.802 (Proportion of true positives among actual positives, aka sensitivity)

F1 Score: 0.641 (Harmonic mean of precision and recall)

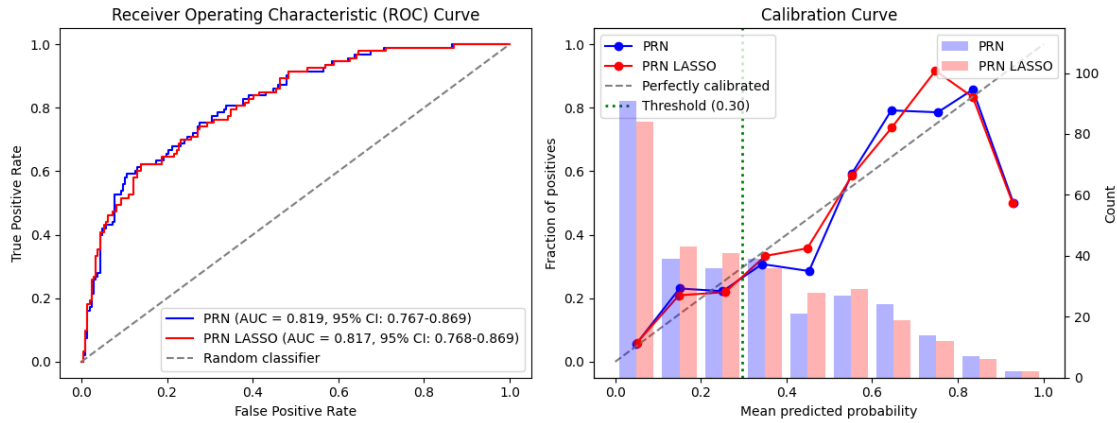


----- Model Performance Comparison -----
Threshold (prevalence) set to: 0.296

PRN:
AUROC: 0.819 (95% CI: 0.767-0.869)
Accuracy: 0.720
Precision: 0.533
Recall: 0.774
F1 Score: 0.632

PRN LASSO:
AUROC: 0.817 (95% CI: 0.768-0.869)
Accuracy: 0.713
Precision: 0.526
Recall: 0.763
F1 Score: 0.623

PRN - PRN LASSO test comparison



----- Model Performance Comparison -----

Threshold (prevalence) set to: 0.296

MLP:

AUROC: 0.808 (95% CI: 0.758-0.856)
 Accuracy: 0.723
 Precision: 0.540
 Recall: 0.720
 F1 Score: 0.618

PRN LASSO:

AUROC: 0.817 (95% CI: 0.763-0.861)
 Accuracy: 0.713
 Precision: 0.526
 Recall: 0.763
 F1 Score: 0.623

MLP - PRN LASSO test comparison

